Machine Learning (CS6L004)

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Lecture 5: Generative Models and Naive Bayes



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Where We Are: Recap of Logistic Regression

In our previous lectures on supervised classification:

- Given dataset $\mathcal{D} = \{(x^{(i)}, y^{(i)})\}_{i=1}^{n}$, we trained a model to output conditional class probabilities directly:

$$P_\theta(y = 1 \mid x) = \sigma(\theta^\top x) = \frac{1}{1 + \exp(-\theta^\top x)}$$

- The model adjusts weights θ to draw a separating hyperplane (decision boundary) between classes.
- We optimized θ via iterative gradient descent by maximizing the conditional log-likelihood.
- We extended this model to Softmax Regression for multi-class classification problem.

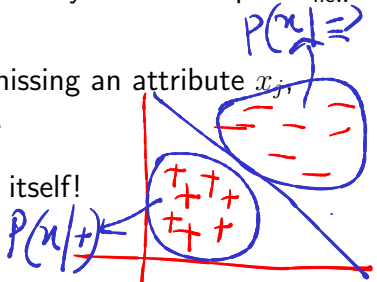
Question: What if our goal goes beyond merely drawing a decision boundary?

Limitations of the Direct Boundary Approach

While Logistic Regression is highly effective, it has clear conceptual and practical limits:

- **No Model of the Input Data:** It only models $P(y \mid x)$ and knows nothing about the distribution of inputs $P(x)$. It cannot tell you what a typical class sample looks like.
- **Cannot Sample/Generate Data:** It cannot generate new synthetic samples x_{new} for a given class.
- **Vulnerable to Missing Features:** If a test sample is missing an attribute x_j, computing $\theta^\top x$ fails directly without explicit imputation.

The Natural Alternative: Model the data-generation story itself!



Part 1: Two Philosophical Approaches to Classification

1. Discriminative Approach

- Focuses strictly on the decision boundary: $P(y \mid x)$.
- Answers: *"Given this specific x, what is the boundary separating class 0 from class 1?"*
- **Examples:** Logistic Regression, Support Vector Machines (SVMs), Artificial Neural Networks (ANN), Decision Trees, Random Forests, Gradient Boosted Trees, k-Nearest Neighbors (k-NN), *etc.*

2. Generative Approach

- Models the complete joint distribution:

$$P(x, y) = P(x \mid y)P(y)$$

- Answers: *"What do features look like when generated by class y, and how common is class y?"*
- **Examples:** Naive Bayes, Gaussian Discriminant Analysis (GDA), Hidden Markov Models (HMM), Gaussian Mixture Models (GMM), Generative Adversarial Networks (GAN), *etc.*

Inference: Understanding Bayes' Rule

Bayes' Rule connects the learned generative model components to the class prediction:

$$\underbrace{P(y = k \mid x)}_{\text{Posterior}} = \frac{\overbrace{P(x \mid y = k)}^{\text{Likelihood}} \cdot \overbrace{P(y = k)}^{\text{Prior}}}{\underbrace{P(x)}_{\text{Evidence / Marginal Likelihood}}}$$

Deconstruction of the Terms:

- **Prior** $P(y = k)$**:** Our initial belief of class k frequency *before* seeing features x.
- **Likelihood** $P(x \mid y = k)$**:** How likely it is that class k produces feature vector x.
- **Evidence** $P(x)$**:** Total probability of observing x, computed as $P(x) = \sum_{l=1}^{K} P(x \mid y = l)P(y = l)$.
- **Posterior** $P(y = k \mid x)$**:** Updated probability that the sample is class k *after* observing x.

Inference: Maximum A Posteriori (MAP) Decision

To make a discrete class prediction for input x, we pick the class with the highest posterior:

$$y_{\text{pred}} = \arg\max_{k \in \{1,\dots,K\}} P(y = k \mid x)$$

$$= \arg\max_{k \in \{1,\dots,K\}} \frac{P(x \mid y = k)P(y = k)}{P(x)}$$

$$= \arg\max_{k \in \{1,\dots,K\}} P(x \mid y = k)P(y = k)$$

- **Why drop $P(x)$?** The evidence term $P(x)$ acts as a common normalization constant and is identical for all classes $k \in \{1, \dots, K\}$.
- **Takeaway:** We only need to compute and compare the numerator (unnormalized joint score) for each class.

Running Problem: Playing Cricket

Goal: Predict whether to **Play Cricket** ($y \in \{\textbf{Yes}, \textbf{No}\}$) given weather conditions.

- **Categorical Features:** $x_1 = \text{Outlook} \in \{\text{Sunny}, \text{Rain}\}$, $x_2 = \text{Wind} \in \{\text{Weak}, \text{Strong}\}$
- **Continuous Feature:** $x_3 = \text{Temperature}\ (^\circ\text{C})$

Day	Outlook (x_1)	Wind (x_2)	Temperature (x_3)	Play Cricket (y)
D1	Sunny	Weak	30°C	No
D2	Sunny	Strong	32°C	No
D3	Sunny	Weak	22°C	Yes
D4	Rain	Weak	20°C	Yes
D5	Rain	Weak	24°C	Yes
D6	Rain	Strong	26°C	No

Query Point to Classify: $x_{\text{new}} = (\text{Outlook} = \text{Sunny},\ \text{Wind} = \text{Weak},\ \text{Temperature} = 24^\circ\text{C})$.

$$P(y=k|x) = \frac{P(x|y=k)\cdot P(y=k)}{P(x)}$$

$$\hookrightarrow P(y=yes|x_{new}) = P(y=yes|\langle x_1=Sunny, x_2=weak, x_3=24^{\circ}C\rangle)$$

Conditional dependency $= \dfrac{P(\langle Sunny, weak, 24\rangle|y=yes)\cdot P(y=yes)}{x_1 \quad x_2 \quad x_3}$

$$\hookrightarrow P(\langle A,B\rangle|C) = P(A|C)\cdot P(B|A,C)$$

A → friend is taking an umbrella

B → The road is wet

C → It is rainy.

Conditional independency

$$P(\langle A,B\rangle|C) = P(A|C)\cdot P(B|C)$$

Part 2: The Naive Bayes Model

The Curse of Dimensionality:

- Let $x = (x_1, x_2, \ldots, x_d)$ be an input with d binary attributes ($x_j \in \{0, 1\}$).
- There are 2^d possible feature combinations for vector x.
- Specifying a full categorical distribution $P(x_1, \ldots, x_d \mid y)$ requires estimating $2^d - 1$ parameters for every class.
- If $d = 100$, $2^{100} \approx 1.26 \times 10^{30}$ parameters (impossible to store or learn from finite data).

The Naive Bayes Assumption:

Given the class label y, all feature attributes $x_1, x_2, \ldots, x_d$ are **conditionally independent**.

$$P(x \mid y = k) = P(x_1, x_2, \ldots, x_d \mid y = k) = \prod_{j=1}^{d} P(x_j \mid y = k)$$

Reduces parameter complexity from exponential $\mathcal{O}(K \cdot 2^d)$ to linear $\mathcal{O}(K \cdot d)$

Why Conditional Independence? (Cricket Dataset)

To evaluate query $x_{\text{new}} = (\text{Sunny}, \text{Weak}, 24°\text{C})$ for class Yes:

Without Independence (Chain Rule):

$$P(x_1 \mid \text{Yes}) \cdot P(x_2 \mid x_1, \text{Yes}) \cdot P(x_3 \mid x_1, x_2, \text{Yes})$$

- $P(\text{Sunny} \mid \text{Yes})$: uses all 3 Yes days (D_3, D_4, D_5).
- $P(\text{Weak} \mid \text{Sunny}, \text{Yes})$: filters to $(\text{Sunny} \wedge \text{Yes}) \rightarrow$ **only** D_3 **(**$N = 1$**)**.
- $P(24°\text{C} \mid \text{Sunny}, \text{Weak}, \text{Yes})$: filters to all three conditions $\rightarrow$ **only** D_3 **(**$N = 1$**)**.
- **Fatal Breakdown:** Fitting Gaussian on $N = 1$ gives variance $\sigma^2 = 0$. Model breaks!

With Independence (Naive Bayes):

$$P(x_1 \mid \text{Yes}) \cdot P(x_2 \mid \text{Yes}) \cdot P(x_3 \mid \text{Yes})$$

- Features condition **only** on class (Yes).
- Every feature uses **all 3 Yes samples**:
- Outlook: uses $\{D_3, D_4, D_5\} \implies P(\text{Sunny} \mid \text{Yes}) = \frac{1}{3}$.
- Wind: uses $\{D_3, D_4, D_5\} \implies P(\text{Weak} \mid \text{Yes}) = 1$.
- Temp: fits $\mathcal{N}(\mu = 22, \sigma^2 = 2.67)$ on all 3.
- **Result:** Stable, robust estimation.

Takeaway: Without independence, data fractures rapidly ($N \rightarrow 1$ or 0). Naive Bayes lets every feature leverage the full class sample size.

Inference: Full Formulation for $P(y \mid x_{\text{new}})$

For query $x_{\text{new}} = (x_1 = \text{Sunny},\ x_2 = \text{Weak},\ x_3 = 24°\text{C})$:

1. **Posterior for Class Yes ($y = \text{Yes}$):**

$$P(\text{Yes} \mid x_{\text{new}}) = \frac{P(x_1 = \text{Sunny} \mid \text{Yes}) \cdot P(x_2 = \text{Weak} \mid \text{Yes}) \cdot P(x_3 = 24 \mid \text{Yes}) \cdot P(\text{Yes})}{P(x_{\text{new}})}$$

2. **Posterior for Class No ($y = \text{No}$):**

$$P(\text{No} \mid x_{\text{new}}) = \frac{P(x_1 = \text{Sunny} \mid \text{No}) \cdot P(x_2 = \text{Weak} \mid \text{No}) \cdot P(x_3 = 24 \mid \text{No}) \cdot P(\text{No})}{P(x_{\text{new}})}$$

Where the Marginal Likelihood (Evidence) $P(x_{\text{new}})$ is the Normalizer:

$$P(x_{\text{new}}) = \underbrace{P(\text{Yes}) \cdot \prod_{j=1}^{3} P(x_j \mid \text{Yes})}_{\text{Score(Yes)}} + \underbrace{P(\text{No}) \cdot \prod_{j=1}^{3} P(x_j \mid \text{No})}_{\text{Score(No)}}$$

Parameter Learning via Closed-Form Counting

Theoretical Formulation (MLE):

1. **Class Priors ($P(y = k) = \phi_k$):**

$$\phi_k = \frac{n_k}{n} = \frac{\text{Count}(y = k)}{n}$$

2. **Categorical Likelihoods:**

$$P(x_j = v \mid y = k) = \frac{\text{Count}(x_j = v, y = k)}{n_k}$$

Requires no gradient descent; computed directly from sample frequencies.

Dataset ($n = 6, n_{\text{Yes}} = 3, n_{\text{No}} = 3$):

1. Class Priors:

$$P(\text{Yes}) = \frac{3}{6} = 0.5, \quad P(\text{No}) = \frac{3}{6} = 0.5$$

2. Categorical Counts for x_{new}:

- **Outlook (x_1):**
  - $P(\text{Sunny} \mid \text{Yes}) = \frac{1}{3}$
  - $P(\text{Sunny} \mid \text{No}) = \frac{2}{3}$
- **Wind (x_2):**
  - $P(\text{Weak} \mid \text{Yes}) = \frac{3}{3} = 1.0$
  - $P(\text{Weak} \mid \text{No}) = \frac{1}{3}$

Notice: $P(\text{Strong} \mid \text{Yes}) = \frac{0}{3} = 0$. What happens if a query has Wind = Strong for class Yes?

The Zero-Probability Problem & Laplace Smoothing

The Issue with Rare / Unseen Attributes:
If a feature value v never appeared in class k:

$$P(x_j = v \mid y = k) = 0$$

$$\implies \prod_{j=1}^{d} P(x_j \mid y = k) = 0$$

A single zero wipes out the entire product!

Solution — Laplace Smoothing: Add pseudo-count $\alpha = 1$, adjust denominator by categories V_j for x_j feature:

$$P(x_j = v \mid y = k) = \frac{\text{Count}(x_j = v, y = k) + 1}{n_k + V_j}$$

Applying to Cricket ($V_1 = V_2 = 2$ categories, $n_{\text{Yes}} = n_{\text{No}} = 3$):

For Class Yes ($N_{\text{Yes}} = 3$):

- $P(\text{Sunny} \mid \text{Yes}) = \frac{1+1}{3+2} = \frac{2}{5} = \mathbf{0.40}$
- $P(\text{Weak} \mid \text{Yes}) = \frac{3+1}{3+2} = \frac{4}{5} = \mathbf{0.80}$

For Class No ($N_{\text{No}} = 3$):

- $P(\text{Sunny} \mid \text{No}) = \frac{2+1}{3+2} = \frac{3}{5} = \mathbf{0.60}$
- $P(\text{Weak} \mid \text{No}) = \frac{1+1}{3+2} = \frac{2}{5} = \mathbf{0.40}$

Part 3: Continuous Features (Gaussian Naive Bayes)

Gaussian Naive Bayes Assumption: For continuous $x_j \in \mathbb{R}$, assume feature within class k is drawn from a 1D Normal distribution $\mathcal{N}(\mu, \sigma^2)$:

$$P(x_j \mid y = k) = \frac{1}{\sqrt{2\pi\sigma_{jk}^2}} \exp\left(-\frac{(x_j - \mu_{jk})^2}{2\sigma_{jk}^2}\right)$$

Parameters computed directly from class k samples:

$$\mu_{jk} = \frac{1}{n_k} \sum_{i:y^{(i)}=k} x_j^{(i)}$$

$$\sigma_{jk}^2 = \frac{1}{n_k} \sum_{i:y^{(i)}=k} (x_j^{(i)} - \mu_{jk})^2$$

Cricket Dataset: Temperature (x_3)

Yes Class $\{22, 20, 24\}$:

- $\mu_{\text{Yes}} = 22.0°\text{C}, \quad \sigma^2_{\text{Yes}} = \frac{8}{3} \approx \mathbf{2.67}$

No Class $\{30, 32, 26\}$:

- $\mu_{\text{No}} = 29.33°\text{C}, \quad \sigma^2_{\text{No}} = \frac{56}{9} \approx \mathbf{6.22}$

Density at Query $x_3 = 24°$C:

- $P(T = 24 \mid \text{Yes}) = \frac{1}{\sqrt{2\pi(2.67)}} e^{-\frac{(24-22)^2}{2(2.67)}} \approx \mathbf{0.1152}$
- $P(T = 24 \mid \text{No}) = \frac{1}{\sqrt{2\pi(6.22)}} e^{-\frac{(24-29.33)^2}{2(6.22)}} \approx \mathbf{0.0163}$

Worked Example: Joint Scores & MAP Decision

Compute Joint Unnormalized Scores for $x_{\text{new}} = (\textbf{Sunny}, \textbf{Weak}, 24°\textbf{C})$:

$$\begin{aligned}\text{Score}(\text{Yes}) &= P(\text{Yes}) \cdot P(\text{Sunny} \mid \text{Yes}) \cdot P(\text{Weak} \mid \text{Yes}) \cdot P(T = 24 \mid \text{Yes}) \\ &= 0.5 \times 0.40 \times 0.80 \times 0.1152 = \mathbf{0.01843}\end{aligned}$$

$$\begin{aligned}\text{Score}(\text{No}) &= P(\text{No}) \cdot P(\text{Sunny} \mid \text{No}) \cdot P(\text{Weak} \mid \text{No}) \cdot P(T = 24 \mid \text{No}) \\ &= 0.5 \times 0.60 \times 0.40 \times 0.0163 = \mathbf{0.00196}\end{aligned}$$

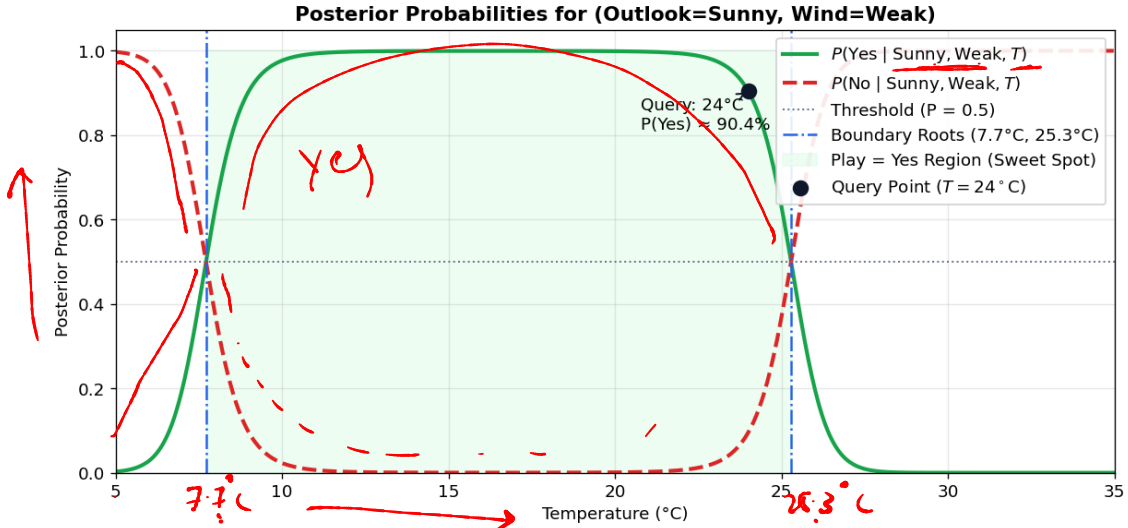
Normalize Scores into Class Posterior Probabilities:

$$P(\text{Play} = \text{Yes} \mid x_{\text{new}}) = \frac{\text{Score}(\text{Yes})}{\text{Score}(\text{Yes}) + \text{Score}(\text{No})} = \frac{0.01843}{0.01843 + 0.00196} \approx \mathbf{90.39\%}$$

$$P(\text{Play} = \text{No} \mid x_{\text{new}}) = \frac{\text{Score}(\text{No})}{\text{Score}(\text{Yes}) + \text{Score}(\text{No})} = \frac{0.00196}{0.01843 + 0.00196} \approx \mathbf{9.61\%}$$

Final Decision: Since $P(\text{Yes} \mid x_{\text{new}}) = \mathbf{90.39\%} > \mathbf{9.61\%} = P(\text{No} \mid x_{\text{new}})$, predict: **Play Cricket = Yes**.

Posterior Probabilities for (Outlook=Sunny, Wind=Weak)



Decision Boundary: Linearity Criterion

1. Log-Posterior Odds Formulation: The boundary between classes occurs where the posterior log-odds ratio equals zero:

$$\log \frac{P(\text{Yes} \mid x)}{P(\text{No} \mid x)} = \underbrace{\log \frac{P(\text{Yes})}{P(\text{No})}}_{\text{Priors}} + \underbrace{\sum_{j=1}^{2} \log \frac{P(x_j \mid \text{Yes})}{P(x_j \mid \text{No})}}_{\text{Discrete } (x_1, x_2)} + \underbrace{\log \frac{P(x_3 \mid \text{Yes})}{P(x_3 \mid \text{No})}}_{\text{Gaussian } (x_3 = T)} = 0$$

Categorical Contribution (x_1, x_2):

- Observed categories contribute additive scalar weights $w_j . x_j$ via counting tables.
- In vector form, this is strictly **first-order**:

$$\sum_{j=1}^{2} \log \frac{P(x_j \mid \text{Yes})}{P(x_j \mid \text{No})} = \mathbf{w}_{\text{cat}}^{\top} \mathbf{x}_{\text{cat}}$$

- Acts purely as an additive **linear shift (bias)**.

Gaussian Continuous Feature ($x_3 = T$):

- Subtracting the log-normal densities yields:

$$a_3 T^2 + w_3 T + C_3 = 0$$

- The quadratic T^2 coefficient is:

$$a_3 = \frac{1}{2\sigma_{\text{No}}^2} - \frac{1}{2\sigma_{\text{Yes}}^2}$$

- Depends entirely on class-conditional variances.

Linearity Rule: If $\sigma_{\text{Yes}}^2 = \sigma_{\text{No}}^2 \implies a_3 = 0$ (**Linear**).
If $\sigma_{\text{Yes}}^2 \neq \sigma_{\text{No}}^2 \implies a_3 \neq 0$ (**Non-Linear / Quadratic**).

Decision Boundary: Application to Cricket Dataset

2. Substituting Parameters from the Cricket Model:

Calculated Parameters:

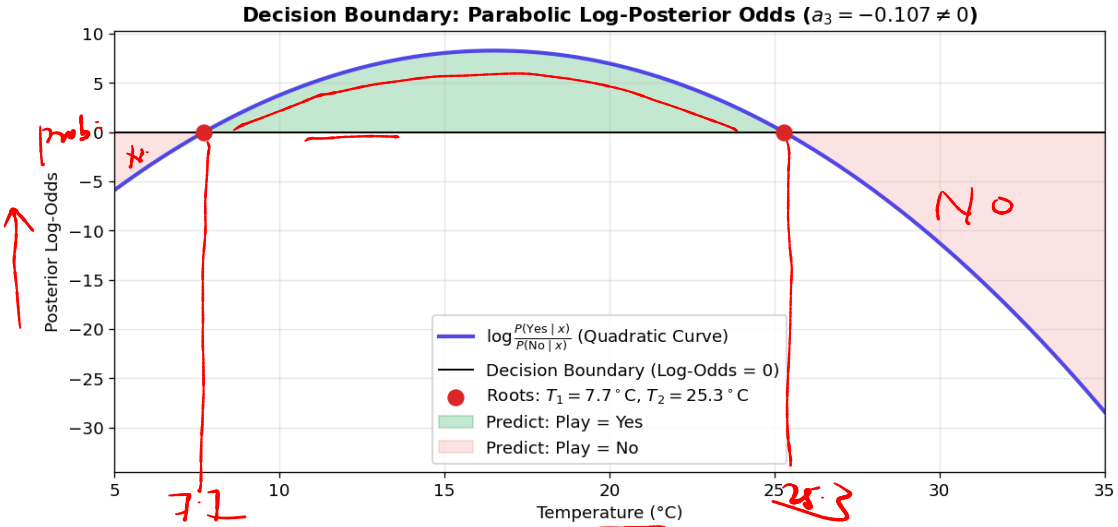
- **Equal Priors:** $\log \frac{0.50}{0.50} = \mathbf{0}$
- **Discrete Shift for (Sunny, Weak):**
$$w_{\text{cat}} = \log \frac{0.40}{0.60} + \log \frac{0.80}{0.40} = +\mathbf{0.288}$$
- **Unequal Temperature Variances:**
$$\sigma^2_{\text{Yes}} = \mathbf{2.67} \neq \mathbf{6.22} = \sigma^2_{\text{No}}$$
- **Surviving Quadratic Coefficient ($a_3 \neq 0$):**
$$a_3 = \frac{1}{2(6.22)} - \frac{1}{2(2.67)} = -\mathbf{0.107}$$

Resulting Boundary in T:

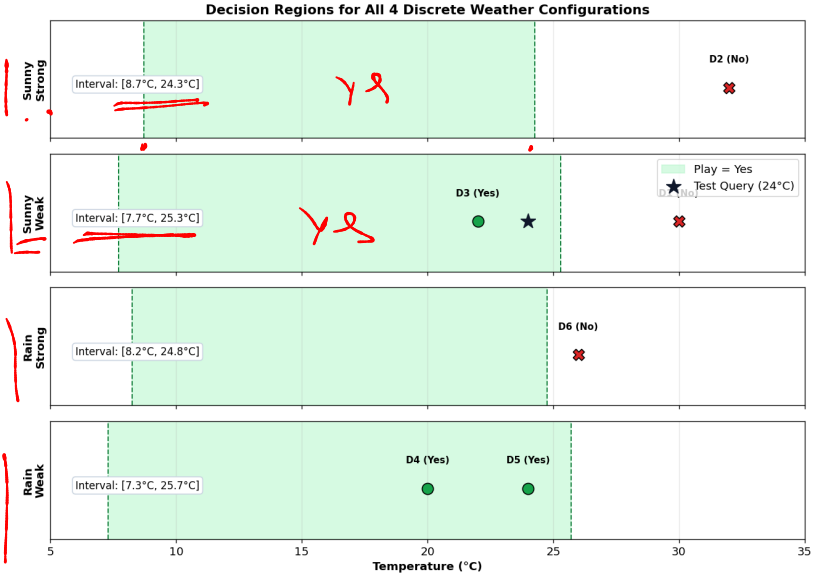
- Combining all terms yields the quadratic inequality:
$$-0.107\,T^2 + 3.524\,T - 20.781 > 0$$
- Solving for the real roots gives an **interval**:
$$\mathbf{7.7}^\circ\text{C} < \text{Temperature} < \mathbf{25.3}^\circ\text{C}$$
- **Geometry:** Rather than a one-sided half-space ($T > T_0$), the model isolates a comfortable **"sweet spot"**.

Final Verdict: Because $\sigma^2_{\text{Yes}} \neq \sigma^2_{\text{No}}$, the model draws a **Non-Linear (Quadratic)** boundary!

Decision Boundary: Parabolic Log-Posterior Odds



Decision Regions for All 4 Discrete Weather Configurations



Part 4: Real-World Application: Text Classification

- Naive Bayes gained prominence through its early success in **Email Spam Filtering**.
- **Question:** How do we convert raw sentences of varying lengths into fixed feature vectors x?

The Bag-of-Words (BoW) Representation:

1. Build a fixed vocabulary $V = \{\text{word}_1, \text{word}_2, \ldots, \text{word}_d\}$ of size $d = |V|$.
2. Represent document x as a binary vector $\phi(x) \in \{0, 1\}^d$:

$$\phi(x)_j = \begin{cases} 1 & \text{if word } j \text{ appears in document } x \\ 0 & \text{otherwise} \end{cases}$$

Bag-of-Words Vector Example

Vocabulary (V):

$$V = \begin{pmatrix} \text{church} \\ \text{doctor} \\ \text{fervently} \\ \text{purple} \\ \text{fast} \end{pmatrix}$$

Document (x): *"The doctor ran fast."*

$$\phi(x) = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 1 \end{pmatrix} \begin{matrix} \leftarrow \text{no "church"} \\ \leftarrow \textbf{contains "doctor"} \\ \leftarrow \text{no "fervently"} \\ \leftarrow \text{no "purple"} \\ \leftarrow \textbf{contains "fast"} \end{matrix}$$

Standard NLP Preprocessing Steps:

- **Stemming:** Reduce words to root forms (e.g., "running", "runs" $\rightarrow$ "run").
- **Stopword Removal:** Filter out uninformative words ("the", "is", "at").

Bernoulli Naive Bayes for Text

For binary text vectors $x \in \{0,1\}^d$, each word presence is modeled as a Bernoulli distribution with Laplace smoothing:

$$P(y = k) = \phi_k \quad \text{(Class Prior)}$$

$$P(x_j = 1 \mid y = k) = \psi_{jk} = \frac{n_{jk} + 1}{n_k + 2} \quad \text{(Smoothed Word Likelihood)}$$

$$P(x_j = 0 \mid y = k) = 1 - \psi_{jk}$$

Document Joint Likelihood:

$$P(x \mid y = k) = \prod_{j=1}^{d} (\psi_{jk})^{x_j} (1 - \psi_{jk})^{1 - x_j}$$

Part 5: Comparison Summary

Property	Discriminative (e.g., Logistic Reg.)	Generative (e.g., Naive Bayes)
Goal	Model $P(y \mid x)$ directly	Model $P(x, y) = P(x \mid y)P(y)$
Training	Iterative optimization (GD)	Closed-form counting / statistics
Speed	Slower on large vocabularies	Ultra-fast $\mathcal{O}(n \cdot d)$ training
Missing Features	Difficult to handle directly	Simple: marginalize out missing x_j
Asymptotic Error	Lower error when data is abundant	Higher bias due to independence

Key Takeaways

- **Generative Modeling:** Shifts focus from merely drawing boundaries to modeling the joint data distribution $P(x, y) = P(x \mid y)P(y)$.
- **Naive Bayes Assumption:** Conditionally independent features factor the high-dimensional likelihood into simple 1D products.
- **Laplace Smoothing:** Critical for discrete/text features to prevent zero-probability errors on unseen tokens.
- **Continuous Data:** Readily handled by fitting 1D Gaussians per feature (Gaussian Naive Bayes).

Next Lecture: Model Evaluation, Generalization, and Regularization.

Few slides adapted from course materials available at:

https://kuleshov-group.github.io/aml-book/intro.html